

Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

Aidan N. Gomez*[†]
University of Toronto
aidan@cs.toronto.edu

Łukasz Kaiser*
Google Brain
lukaszkaiser@google.com

Illia Polosukhin*[‡]
illia.polosukhin@gmail.com

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

1 Introduction

Recurrent neural networks, long short-term memory [12] and gated recurrent [7] neural networks in particular, have been firmly established as state of the art approaches in sequence modeling and transduction problems such as language modeling and machine translation [29, 2, 5]. Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [31, 21, 13].

*Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

[†]Work performed while at Google Brain.

[‡]Work performed while at Google Research.

注意力就是你所需要的一切

Ashish Vaswani*
Google Brain
avaswani@google.com

诺姆·沙泽尔*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google
Researchusz@google.com

Llion Jones*
Google Research
llion@google.com

艾丹·N·戈麦斯*[†]多伦多大学
aidan@cs.toronto.edu

卢卡什·凯泽*Google Brain
lukaszkaiser@google.com

伊利亚·波洛苏欣*[‡]
illia.polosukhin@gmail.com

摘要

主流的序列转导模型基于复杂的循环或卷积神经网络，其中包含编码器和解码器。性能最佳的模型还通过注意力机制将编码器与解码器连接起来。我们提出一种全新的简单网络架构——Transformer，它完全基于注意力机制，彻底摒弃了循环和卷积。在两个机器翻译任务上的实验表明，这些模型在质量上更优，同时具有更高的并行度，并且所需的训练时间显著减少。我们的模型在 WMT 2014 英语到德语翻译任务上取得了 28.4 的 BLEU 分数，比包括集成在内的现有最佳结果提高了超过 2 个 BLEU。在 WMT 2014 英语到法语翻译任务上，我们的模型在 8 块 GPU 上训练 3.5 天后，取得了新的单模型最先进 BLEU 分数 41.0，这仅是文献中最佳模型训练成本的一小部分。

1 引言

循环神经网络，尤其是长短期记忆 [12] 和门控循环 [7] 神经网络，已被牢固确立为序列建模和转换问题（如语言建模和机器翻译 [29, 2, 5]）中的最优水平方法。此后，众多努力持续推动着循环语言模型和编码器-解码器架构 [31, 21, 13] 的边界。

*同等贡献。作者顺序随机排列。Jakob 提出用自注意力替代循环神经网络，并启动了对这一想法的评估工作。Ashish 与 Illia 一起设计并实现了最初的 Transformer 模型，并深度参与了这项工作的方方面面。Noam 提出了缩放点积注意力、多头注意力和无参数位置表示，并成为另一个几乎参与每个细节的人。Niki 在我们最初的代码库和 tensor2tensor 中设计、实现、调优并评估了无数模型变体。Llion 也尝试了新的模型变体，负责了我们最初的代码库，以及高效推理和可视化。Lukasz 和 Aidan 花费了无数个漫长的日夜来设计并实现 tensor2tensor 的各个部分，替换了我们早期的代码库，极大地改善了结果并大幅加速了我们的研究。
[†]在 Google Brain 期间完成的工作。
[‡]在 Google Research 期间完成的工作。

Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states h_t , as a function of the previous hidden state h_{t-1} and the input for position t . This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples. Recent work has achieved significant improvements in computational efficiency through factorization tricks [18] and conditional computation [26], while also improving model performance in case of the latter. The fundamental constraint of sequential computation, however, remains.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2, 16]. In all but a few cases [22], however, such attention mechanisms are used in conjunction with a recurrent network.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

2 Background

The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU [20], ByteNet [15] and ConvS2S [8], all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions. In these models, the number of operations required to relate signals from two arbitrary input or output positions grows in the distance between positions, linearly for ConvS2S and logarithmically for ByteNet. This makes it more difficult to learn dependencies between distant positions [11]. In the Transformer this is reduced to a constant number of operations, albeit at the cost of reduced effective resolution due to averaging attention-weighted positions, an effect we counteract with Multi-Head Attention as described in section 3.2.

Self-attention, sometimes called intra-attention is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence. Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4, 22, 23, 19].

End-to-end memory networks are based on a recurrent attention mechanism instead of sequence-aligned recurrence and have been shown to perform well on simple-language question answering and language modeling tasks [28].

To the best of our knowledge, however, the Transformer is the first transduction model relying entirely on self-attention to compute representations of its input and output without using sequence-aligned RNNs or convolution. In the following sections, we will describe the Transformer, motivate self-attention and discuss its advantages over models such as [14, 15] and [8].

3 Model Architecture

Most competitive neural sequence transduction models have an encoder-decoder structure [5, 2, 29]. Here, the encoder maps an input sequence of symbol representations $(x_1, \dots, x_n)$ to a sequence of continuous representations $\mathbf{z} = (z_1, \dots, z_n)$. Given $\mathbf{z}$, the decoder then generates an output sequence $(y_1, \dots, y_m)$ of symbols one element at a time. At each step the model is auto-regressive [9], consuming the previously generated symbols as additional input when generating the next.

The Transformer follows this overall architecture using stacked self-attention and point-wise, fully connected layers for both the encoder and decoder, shown in the left and right halves of Figure 1, respectively.

3.1 Encoder and Decoder Stacks

Encoder: The encoder is composed of a stack of $N = 6$ identical layers. Each layer has two sub-layers. The first is a multi-head self-attention mechanism, and the second is a simple, position-

循环模型通常沿着输入和输出序列的符号位置分解计算。将位置与计算时间中的步数对齐后，它们会生成一系列隐藏状态 h_t ，作为前一隐藏状态 h_{t-1} 和位置 t 的输入的函数。这种固有的顺序性使得训练样本内无法并行化，而这在序列长度较长时变得至关重要，因为记忆限制会约束跨样本的批处理。近期工作通过分解技巧 [18] 和条件计算 [26]，在计算效率方面实现了显著提升，同时后者还改善了模型性能。然而，顺序计算这一根本限制仍然存在。

注意力机制已成为各种任务中引人注目的序列建模和转换模型不可或缺的组成部分，它允许对依赖关系进行建模，而无需考虑它们在输入或输出序列中的距离 [2, 16]。然而，在几乎所有情况下 [22]，这种注意力机制都是与循环网络结合使用的。

在这项工作中，我们提出了Transformer，这是一种摒弃循环而完全依赖注意力机制来绘制输入和输出之间全局依赖关系的模型架构。Transformer允许显著提高并行化程度，并且在八个P100 GPU上仅训练十二小时后，就能在翻译质量上达到新的最优水平。

2 背景

减少顺序计算的目标也构成了扩展神经GPU [20]、ByteNet [15] 和ConvS2S [8] 的基础，它们都使用卷积神经网络作为基本构建块，为所有输入和输出位置并行计算隐藏表示。在这些模型中，关联来自两个任意输入或输出位置的信号所需的运算次数随着位置之间的距离而增长，对于ConvS2S是线性的，对于ByteNet是对数的。这使得学习远距离位置之间的依赖关系变得更加困难 [11]。在Transformer中，这被减少到恒定数量的运算，尽管代价是由于对注意力加权位置进行平均而降低了有效分辨率，我们通过第3.2节中描述的多头注意力来抵消这种影响。

自注意力，有时称为内部注意力，是一种将单个序列中不同位置关联起来以计算该序列表示的注意力机制。自注意力已成功应用于多种任务，包括阅读理解、抽象式摘要、文本蕴涵以及学习与任务无关的句子表示 [4, 22, 23, 19]。

端到端记忆网络基于循环注意力机制而非序列对齐循环，并且已被证明在简单语言问答和语言建模任务上表现良好 [28]。

然而，据我们所知，Transformer是第一个完全依赖自注意力来计算其输入和输出表示而不使用序列对齐循环神经网络或卷积的转导模型。在接下来的章节中，我们将描述Transformer，阐释自注意力的动机，并讨论其相较于 [14, 15] 和 [8]等模型的优势。

3 模型架构

大多数具有竞争力的神经序列转换模型都采用编码器-解码器结构 [5, 2, 29]。其中，编码器将输入符号表示序列 $(x_1, \dots, x_n)$ 映射为连续表示序列 $\mathbf{z} = (z_1, \dots, z_n)$ 。给定 $\mathbf{z}$ 后，解码器随后逐元素生成输出符号序列 $(y_1, \dots, y_m)$ 。模型在每一步中都是自回归的[9]，在生成下一个元素时会将先前生成的符号作为额外输入。Transformer 遵循这一整体架构，编码器和解码器均使用堆叠的自注意力和逐点全连接层，分别如图1的左半部分和右半部分所示。

3.1 编码器和解码器堆栈

编码器：编码器由一组 $N = 6$ 相同层堆叠而成。每一层有两个子层。第一个子层是多头自注意力机制，第二个子层是简单的、位置-

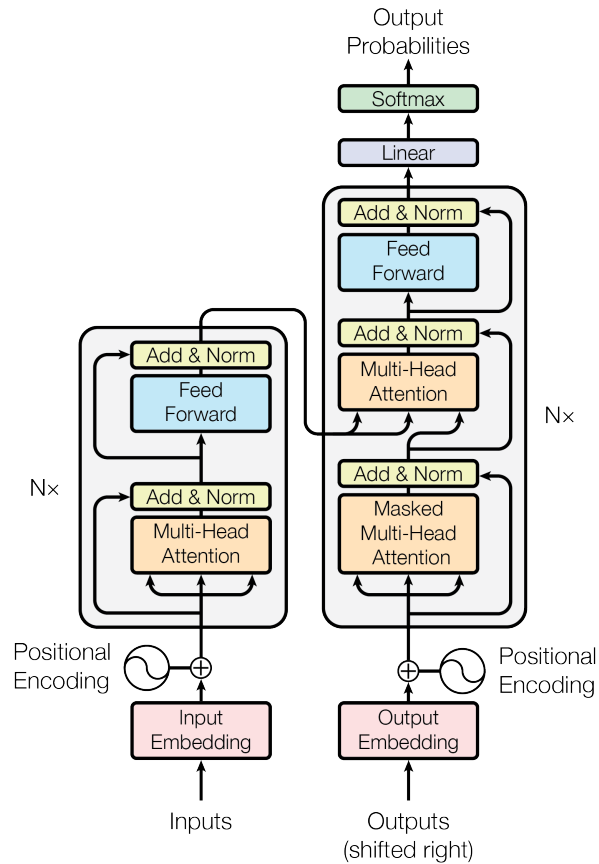


Figure 1: The Transformer - model architecture.

wise fully connected feed-forward network. We employ a residual connection [10] around each of the two sub-layers, followed by layer normalization [1]. That is, the output of each sub-layer is $\text{LayerNorm}(x + \text{Sublayer}(x))$, where $\text{Sublayer}(x)$ is the function implemented by the sub-layer itself. To facilitate these residual connections, all sub-layers in the model, as well as the embedding layers, produce outputs of dimension $d_{\text{model}} = 512$.

Decoder: The decoder is also composed of a stack of $N = 6$ identical layers. In addition to the two sub-layers in each encoder layer, the decoder inserts a third sub-layer, which performs multi-head attention over the output of the encoder stack. Similar to the encoder, we employ residual connections around each of the sub-layers, followed by layer normalization. We also modify the self-attention sub-layer in the decoder stack to prevent positions from attending to subsequent positions. This masking, combined with fact that the output embeddings are offset by one position, ensures that the predictions for position i can depend only on the known outputs at positions less than i .

3.2 Attention

An attention function can be described as mapping a query and a set of key-value pairs to an output, where the query, keys, values, and output are all vectors. The output is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key.

3.2.1 Scaled Dot-Product Attention

We call our particular attention "Scaled Dot-Product Attention" (Figure 2). The input consists of queries and keys of dimension d_k , and values of dimension d_v . We compute the dot products of the

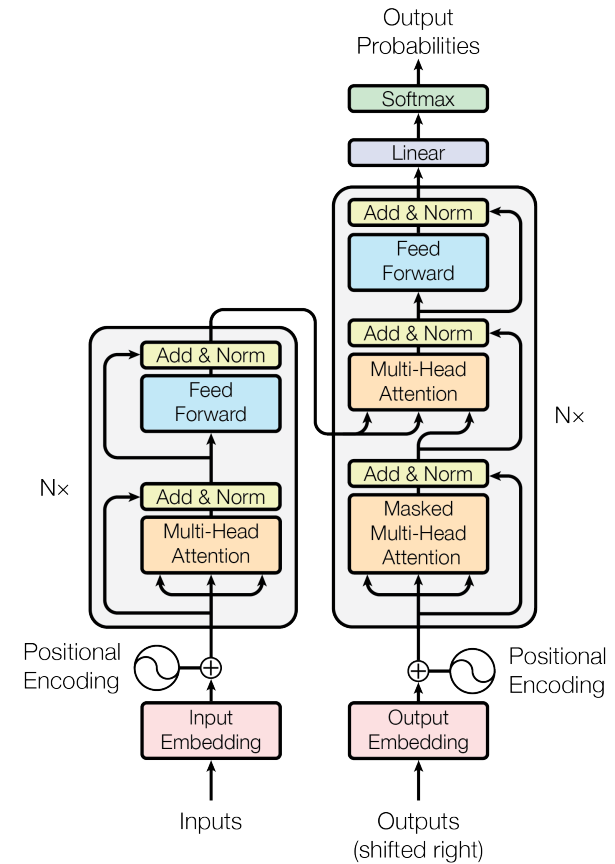


图1: Transformer——模型架构。

逐位置的全连接前馈网络。我们在两个子层的每一个周围采用残差连接 [10]，随后进行层归一化 [1]。也就是说，每个子层的输出为 $\text{LayerNorm}(x + \text{Sublayer}(x))$ ，其中 $\text{Sublayer}(x)$ 是该子层自身实现的函数。为便于这些残差连接，模型中的所有子层以及嵌入层都产生维度为 $d_{\text{model}} = 512$ 的输出。

解码器：解码器同样由 $N = 6$ 相同层的堆栈组成。除了每个编码器层中的两个子层之外，解码器还插入了第三个子层，该子层对编码器堆栈的输出执行多头注意力。与编码器类似，我们在每个子层周围采用残差连接，其后紧跟层归一化。我们还修了解码器堆栈中的自注意力子层，以防止位置关注到后续位置。这种掩码与输出嵌入偏移一个位置的事实相结合，确保了位置 i 的预测只能依赖于小于 i 的位置上的已知输出。

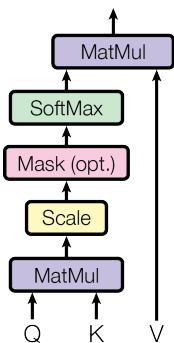
3.2 注意力

注意力函数可以描述为将一个查询和一组键值对映射到一个输出，其中查询、键、值和输出都是向量。输出计算为值的加权和，其中分配给每个值的权重由查询与相应键的兼容性函数计算得出。

3.2.1 缩放点积注意力

我们将这种特别的注意力称为“缩放点积注意力”（图2）。输入由维度为 d_k 的查询和键，以及维度为 d_v 的值组成。我们计算

Scaled Dot-Product Attention



Multi-Head Attention

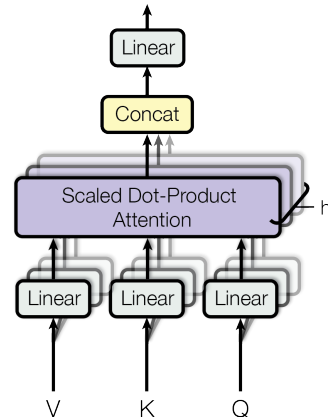


Figure 2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.

query with all keys, divide each by $\sqrt{d_k}$, and apply a softmax function to obtain the weights on the values.

In practice, we compute the attention function on a set of queries simultaneously, packed together into a matrix Q . The keys and values are also packed together into matrices K and V . We compute the matrix of outputs as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

The two most commonly used attention functions are additive attention [2], and dot-product (multiplicative) attention. Dot-product attention is identical to our algorithm, except for the scaling factor of $\frac{1}{\sqrt{d_k}}$. Additive attention computes the compatibility function using a feed-forward network with a single hidden layer. While the two are similar in theoretical complexity, dot-product attention is much faster and more space-efficient in practice, since it can be implemented using highly optimized matrix multiplication code.

While for small values of d_k the two mechanisms perform similarly, additive attention outperforms dot product attention without scaling for larger values of d_k [3]. We suspect that for large values of d_k , the dot products grow large in magnitude, pushing the softmax function into regions where it has extremely small gradients⁴. To counteract this effect, we scale the dot products by $\frac{1}{\sqrt{d_k}}$.

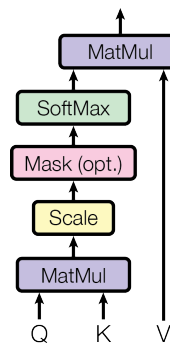
3.2.2 Multi-Head Attention

Instead of performing a single attention function with d_{model} -dimensional keys, values and queries, we found it beneficial to linearly project the queries, keys and values h times with different, learned linear projections to d_k , d_k and d_v dimensions, respectively. On each of these projected versions of queries, keys and values we then perform the attention function in parallel, yielding d_v -dimensional output values. These are concatenated and once again projected, resulting in the final values, as depicted in Figure 2.

Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions. With a single attention head, averaging inhibits this.

⁴To illustrate why the dot products get large, assume that the components of q and k are independent random variables with mean 0 and variance 1. Then their dot product, $q \cdot k = \sum_{i=1}^{d_k} q_i k_i$, has mean 0 and variance d_k .

缩放点积注意力



多头注意力

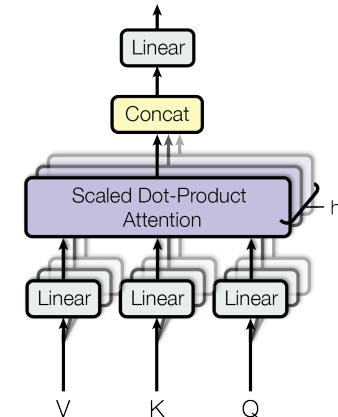


图2：（左）缩放点积注意力。（右）多头注意力由若干并行运行的注意力层组成。

查询与所有键的点积，将每个点积除以 $\sqrt{d_k}$ ，并应用softmax函数来获得值的权重。

在实践中，我们同时在一组查询上计算注意力函数，将它们打包在一起形成一个矩阵 Q 。键和值也被打包在一起形成矩阵 K 和 V 。我们按如下方式计算输出矩阵：

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

两种最常用的注意力函数是加性注意力 [2], 和点积（乘法）注意力。点积注意力与我们的算法相同，只是少了缩放因子 $\frac{1}{\sqrt{d_k}}$ 。加性注意力使用具有单个隐藏层的前馈网络来计算兼容性函数。虽然两者在理论复杂度上相似，但点积注意力在实践中更快、更节省空间，因为它可以使用高度优化的矩阵乘法代码来实现。

虽然对于较小的 d_k 值，两种机制表现相似，但在没有缩放的情况下，对于较大的 d_k [3]值，加性注意力优于点积注意力。我们怀疑对于较大的 d_k 值，点积在数量级上会变得很大，从而将softmax函数推到梯度⁴极小的区域。为了抵消这种影响，我们将点积按 $\frac{1}{\sqrt{d_k}}$ 进行缩放。

3.2.2 多头注意力

与使用 d_{model} 维的键、值和查询来执行单个注意力函数不同，我们发现将查询、键和值分别用不同的、可学习的线性投影进行 h 次投影，得到 d_k 、 d_k 和 d_v 维是有益的。然后，在这些投影后的查询、键和值的每个版本上，我们并行执行注意力函数，得到 d_v 维的输出值。这些值被拼接起来并再次进行投影，得到最终的值，如图2所示。

多头注意力允许模型在不同位置联合关注来自不同表示子空间的信息。而使用单个注意力头时，平均会抑制这一点。

⁴为了说明为什么点积会变大，假设 q 和 k 的分量是均值为0、方差为1的独立随机变量。那么它们的点积 $q \cdot k = \sum_{i=1}^{d_k} q_i k_i$ 的均值为0，方差为 d_k 。

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

$$\text{where head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

Where the projections are parameter matrices $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ and $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$.

In this work we employ $h = 8$ parallel attention layers, or heads. For each of these we use $d_k = d_v = d_{\text{model}}/h = 64$. Due to the reduced dimension of each head, the total computational cost is similar to that of single-head attention with full dimensionality.

3.2.3 Applications of Attention in our Model

The Transformer uses multi-head attention in three different ways:

- In "encoder-decoder attention" layers, the queries come from the previous decoder layer, and the memory keys and values come from the output of the encoder. This allows every position in the decoder to attend over all positions in the input sequence. This mimics the typical encoder-decoder attention mechanisms in sequence-to-sequence models such as [31, 2, 8].
- The encoder contains self-attention layers. In a self-attention layer all of the keys, values and queries come from the same place, in this case, the output of the previous layer in the encoder. Each position in the encoder can attend to all positions in the previous layer of the encoder.
- Similarly, self-attention layers in the decoder allow each position in the decoder to attend to all positions in the decoder up to and including that position. We need to prevent leftward information flow in the decoder to preserve the auto-regressive property. We implement this inside of scaled dot-product attention by masking out (setting to $-\infty$) all values in the input of the softmax which correspond to illegal connections. See Figure 2.

3.3 Position-wise Feed-Forward Networks

In addition to attention sub-layers, each of the layers in our encoder and decoder contains a fully connected feed-forward network, which is applied to each position separately and identically. This consists of two linear transformations with a ReLU activation in between.

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \quad (2)$$

While the linear transformations are the same across different positions, they use different parameters from layer to layer. Another way of describing this is as two convolutions with kernel size 1. The dimensionality of input and output is $d_{\text{model}} = 512$, and the inner-layer has dimensionality $d_{ff} = 2048$.

3.4 Embeddings and Softmax

Similarly to other sequence transduction models, we use learned embeddings to convert the input tokens and output tokens to vectors of dimension d_{model} . We also use the usual learned linear transformation and softmax function to convert the decoder output to predicted next-token probabilities. In our model, we share the same weight matrix between the two embedding layers and the pre-softmax linear transformation, similar to [24]. In the embedding layers, we multiply those weights by $\sqrt{d_{\text{model}}}$.

3.5 Positional Encoding

Since our model contains no recurrence and no convolution, in order for the model to make use of the order of the sequence, we must inject some information about the relative or absolute position of the tokens in the sequence. To this end, we add "positional encodings" to the input embeddings at the

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

$$\text{其中 head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

其中投影是参数矩阵 $W_i^Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$ 、 $W_i^K \in \mathbb{R}^{d_{\text{model}} \times d_k}$ 、 $W_i^V \in \mathbb{R}^{d_{\text{model}} \times d_v}$ 和 $W^O \in \mathbb{R}^{hd_v \times d_{\text{model}}}$ 。

在这项工作中，我们使用了 $h = 8$ 个并行的注意力层或头。对于其中每个头，我们使用 $d_k = d_v = d_{\text{model}}/h = 64$ 。由于每个头的维度有所降低，总的计算成本与使用完整维度的单头注意力相近。

3.2.3 注意力在我们模型中的应用

Transformer 以三种不同的方式使用了多头注意力：

- 在“编码器-解码器注意力”层中，查询来自上一个解码器层，而记忆中的键和值来自编码器的输出。这使得解码器中的每个位置都能关注输入序列中的所有位置。这模拟了序列到序列模型（如 [31, 2, 8]）中典型的编码器-解码器注意力机制。
- 编码器包含自注意力层。在自注意力层中，所有的键、值和查询都来自同一个地方，在此情况下是编码器中上一层的输出。编码器中的每个位置都可以关注编码器上一层的所有位置。
- 类似地，解码器中的自注意力层允许解码器中的每个位置关注解码器中直到并包括该位置的所有位置。我们需要防止解码器中的向左信息流，以保持自回归特性。我们通过缩放点积注意力内部进行掩码（将softmax输入中对应于非法连接的所有值设置为 $-\infty$ ）来实现这一点。参见图2。

3.3 位置前馈网络

除了注意力子层之外，我们的编码器和解码器中的每一层还包含一个全连接的前馈网络，该网络分别且相同地应用于每个位置。它由两个线性变换组成，中间有一个ReLU激活。

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2 \quad (2)$$

虽然线性变换在不同位置上是相同的，但它们在不同层之间使用不同的参数。另一种描述方式是将其视为两个卷积核大小为1的卷积。输入和输出的维度为 $d_{\text{model}} = 512$ ，内层的维度为 $d_{ff} = 2048$ 。

3.4 嵌入与 Softmax

与其他序列转导模型类似，我们使用学习嵌入将输入标记和输出标记转换为维度为 d_{model} 的向量。我们还使用常规的学习线性变换和softmax函数将解码器输出转换为预测的下一个标记概率。在我们的模型中，我们在两个嵌入层和softmax前线性变换之间共享相同的权重矩阵，类似于 [24]。在嵌入层中，我们将这些权重乘以 $\sqrt{d_{\text{model}}}$ 。

3.5 位置编码

由于我们的模型不包含循环和卷积，为了让模型能够利用序列的顺序信息，我们必须注入一些关于词元在序列中相对或绝对位置的信息。为此，我们在输入嵌入上添加“位置编码”，

Table 1: Maximum path lengths, per-layer complexity and minimum number of sequential operations for different layer types. n is the sequence length, d is the representation dimension, k is the kernel size of convolutions and r the size of the neighborhood in restricted self-attention.

Layer Type	Complexity per Layer	Sequential Operations	Maximum Path Length
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

bottoms of the encoder and decoder stacks. The positional encodings have the same dimension d_{model} as the embeddings, so that the two can be summed. There are many choices of positional encodings, learned and fixed [8].

In this work, we use sine and cosine functions of different frequencies:

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$

where pos is the position and i is the dimension. That is, each dimension of the positional encoding corresponds to a sinusoid. The wavelengths form a geometric progression from 2π to $10000 \cdot 2\pi$. We chose this function because we hypothesized it would allow the model to easily learn to attend by relative positions, since for any fixed offset k , PE_{pos+k} can be represented as a linear function of PE_{pos} .

We also experimented with using learned positional embeddings [8] instead, and found that the two versions produced nearly identical results (see Table 3 row (E)). We chose the sinusoidal version because it may allow the model to extrapolate to sequence lengths longer than the ones encountered during training.

4 Why Self-Attention

In this section we compare various aspects of self-attention layers to the recurrent and convolutional layers commonly used for mapping one variable-length sequence of symbol representations $(x_1, \dots, x_n)$ to another sequence of equal length $(z_1, \dots, z_n)$, with $x_i, z_i \in \mathbb{R}^d$, such as a hidden layer in a typical sequence transduction encoder or decoder. Motivating our use of self-attention we consider three desiderata.

One is the total computational complexity per layer. Another is the amount of computation that can be parallelized, as measured by the minimum number of sequential operations required.

The third is the path length between long-range dependencies in the network. Learning long-range dependencies is a key challenge in many sequence transduction tasks. One key factor affecting the ability to learn such dependencies is the length of the paths forward and backward signals have to traverse in the network. The shorter these paths between any combination of positions in the input and output sequences, the easier it is to learn long-range dependencies [11]. Hence we also compare the maximum path length between any two input and output positions in networks composed of the different layer types.

As noted in Table 1, a self-attention layer connects all positions with a constant number of sequentially executed operations, whereas a recurrent layer requires $O(n)$ sequential operations. In terms of computational complexity, self-attention layers are faster than recurrent layers when the sequence length n is smaller than the representation dimensionality d , which is most often the case with sentence representations used by state-of-the-art models in machine translations, such as word-piece [31] and byte-pair [25] representations. To improve computational performance for tasks involving very long sequences, self-attention could be restricted to considering only a neighborhood of size r in

表 1: 不同层类型的最大路径长度、每层复杂度和最少顺序操作数。 n 是序列长度, d 是表示维度, k 是卷积的卷积核大小, r 是受限自注意力中的邻域大小。

层类型	每层复杂度	顺序操作数	最大路径长度
自注意力	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
循环	$O(n \cdot d^2)$	$O(n)$	$O(n)$
卷积	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
自注意力 (受限)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

位置编码位于编码器和解码器堆栈的底部。位置编码与嵌入具有相同的维度 d_{model} , 因此二者可以相加。位置编码有多种选择, 包括可学习的和固定的 [8]。

在这项工作中, 我们使用不同频率的正弦和余弦函数:

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{\text{model}}})$$

其中 pos 是位置, i 是维度。也就是说, 位置编码的每个维度都对应一个正弦曲线。其波长构成从 2π 到 $10000 \cdot 2\pi$ 的等比数列。我们选择这个函数, 是因为我们假设它能让模型轻松学会根据相对位置进行注意力计算, 因为对于任何固定偏移量 k , PE_{pos+k} 都可以表示为 PE_{pos} 的线性函数。

我们还尝试改用学习式位置嵌入 [8], 并发现两种版本产生的结果几乎相同 (见表3第(E)行)。我们选择正弦版本, 是因为它可能使模型能够外推到比训练期间遇到的序列长度更长的序列。

4 为什么使用自注意力

在本节中, 我们将自注意力层的各个方面与常用于将一个变长符号表示序列 $(x_1, \dots, x_n)$ 映射到另一个等长序列 $(z_1, \dots, z_n)$ 的循环层和卷积层进行比较, 其中 $x_i, z_i \in \mathbb{R}^d$, 例如典型序列转导编码器或解码器中的隐藏层。为了说明我们使用自注意力的动机, 我们考虑三个需求。

一个是每层总计算复杂度。另一个是可并行化的计算量, 以所需的最少顺序操作数来衡量。

第三个是网络中长程依赖之间的路径长度。学习长程依赖是许多序列转导任务中的关键挑战。影响学习此类依赖能力的一个关键因素是前向和反向信号在网络中必须经过的路径长度。输入和输出序列中任意位置组合之间的这些路径越短, 就越容易学习长程依赖 [11]。因此, 我们还比较了由不同层类型组成的网络中任意两个输入和输出位置之间的最大路径长度。

正如表1所述, 自注意力层通过恒定数量的顺序执行操作连接所有位置, 而循环层需要 $O(n)$ 次顺序操作。在计算复杂度方面, 当序列长度 n 小于表示维度 d 时, 自注意力层比循环层更快, 这在机器翻译中最先进模型使用的句子表示中最为常见, 例如词片段[31] 和字节对 [25] 表示。为了提高涉及非常长序列的任务的计算性能, 可以将自注意力限制为仅考虑大小为 r 的邻域,

the input sequence centered around the respective output position. This would increase the maximum path length to $O(n/r)$. We plan to investigate this approach further in future work.

A single convolutional layer with kernel width $k < n$ does not connect all pairs of input and output positions. Doing so requires a stack of $O(n/k)$ convolutional layers in the case of contiguous kernels, or $O(\log_k(n))$ in the case of dilated convolutions [15], increasing the length of the longest paths between any two positions in the network. Convolutional layers are generally more expensive than recurrent layers, by a factor of k . Separable convolutions [6], however, decrease the complexity considerably, to $O(k \cdot n \cdot d + n \cdot d^2)$. Even with $k = n$, however, the complexity of a separable convolution is equal to the combination of a self-attention layer and a point-wise feed-forward layer, the approach we take in our model.

As side benefit, self-attention could yield more interpretable models. We inspect attention distributions from our models and present and discuss examples in the appendix. Not only do individual attention heads clearly learn to perform different tasks, many appear to exhibit behavior related to the syntactic and semantic structure of the sentences.

5 Training

This section describes the training regime for our models.

5.1 Training Data and Batching

We trained on the standard WMT 2014 English-German dataset consisting of about 4.5 million sentence pairs. Sentences were encoded using byte-pair encoding [3], which has a shared source-target vocabulary of about 37000 tokens. For English-French, we used the significantly larger WMT 2014 English-French dataset consisting of 36M sentences and split tokens into a 32000 word-piece vocabulary [31]. Sentence pairs were batched together by approximate sequence length. Each training batch contained a set of sentence pairs containing approximately 25000 source tokens and 25000 target tokens.

5.2 Hardware and Schedule

We trained our models on one machine with 8 NVIDIA P100 GPUs. For our base models using the hyperparameters described throughout the paper, each training step took about 0.4 seconds. We trained the base models for a total of 100,000 steps or 12 hours. For our big models,(described on the bottom line of table 3), step time was 1.0 seconds. The big models were trained for 300,000 steps (3.5 days).

5.3 Optimizer

We used the Adam optimizer [17] with $\beta_1 = 0.9$, $\beta_2 = 0.98$ and $\epsilon = 10^{-9}$. We varied the learning rate over the course of training, according to the formula:

$$lrate = d_{\text{model}}^{-0.5} \cdot \min(step_num^{-0.5}, step_num \cdot warmup_steps^{-1.5}) \quad (3)$$

This corresponds to increasing the learning rate linearly for the first $warmup_steps$ training steps, and decreasing it thereafter proportionally to the inverse square root of the step number. We used $warmup_steps = 4000$.

5.4 Regularization

We employ three types of regularization during training:

Residual Dropout We apply dropout [27] to the output of each sub-layer, before it is added to the sub-layer input and normalized. In addition, we apply dropout to the sums of the embeddings and the positional encodings in both the encoder and decoder stacks. For the base model, we use a rate of $P_{drop} = 0.1$.

该邻域位于以相应输出位置为中心的输入序列中。这将使最大路径长度增加到 $O(n/r)$ 。我们计划在未来的工作中进一步研究这种方法。

卷积核宽度为 $k < n$ 的单层卷积层无法连接所有的输入和输出位置对。要做到这一点，在连续卷积核的情况下需要堆叠 $O(n/k)$ 层卷积层，在膨胀卷积 [15] 的情况下则需要 $O(\log_k(n))$ 层，这会增加网络中任意两个位置之间最长路径的长度。卷积层通常比循环层更昂贵，代价高出 k 倍。然而，可分离卷积 [6] 可以将复杂度大幅降低至 $O(k \cdot n \cdot d + n \cdot d^2)$ 。但即使采用 $k = n$ ，可分离卷积的复杂度仍等同于自注意力层与逐点前馈层的组合，这正是我们模型中所采用的方法。

作为附带的好处，自注意力可以产生更具可解释性的模型。我们检查了模型中的注意力分布，并在附录中展示和讨论了相关示例。不仅各个注意力头清楚地学会了执行不同的任务，许多注意力头还表现出与句子的句法和语义结构相关的行为。

5 训练

本节描述我们模型的训练机制。

5.1 训练数据与批处理

我们在标准的 WMT 2014 英德数据集上进行了训练，该数据集包含约 450 万个句对。句子使用字节对编码 [3] 进行编码，其共享的源语言-目标语言词汇表约有 37000 个词元。对于英法任务，我们使用了规模显著更大的 WMT 2014 英法数据集，该数据集包含 3600 万个句子，并将词元拆分为一个包含 32000 个词片词汇表 [31]。句对按近似序列长度进行批处理。每个训练批次包含一组句对，其中大约包含 25000 个源语言词元和 25000 个目标语言词元。

5.2 硬件与训练计划

我们在一台配备 8 块 NVIDIA P100 GPU 的机器上训练模型。对于使用全文所述超参数的基础模型，每个训练步大约耗时 0.4 秒。我们训练基础模型共计 100000 步，约 12 小时。对于我们的大模型（见表 3 最后一行所述），每步耗时为 1.0 秒。大模型训练了 300000 步（3.5 天）。

5.3 优化器

我们使用 Adam 优化器 [17]，其中 $\beta_1 = 0.9$ 、 $\beta_2 = 0.98$ 和 $\epsilon = 10^{-9}$ 。我们在训练过程中根据以下公式调整学习率：

$$lrate = d_{\text{model}}^{-0.5} \cdot \min(step_num^{-0.5}, step_num \cdot warmup_steps^{-1.5}) \quad (3)$$

这对应于在前 $warmup_steps$ 个训练步数内线性增加学习率，此后按步数的平方根倒数成比例地降低学习率。我们使用了 $warmup_steps = 4000$ 。

5.4 正则化

我们在训练期间采用三种正则化：

残差丢弃 我们在每个子层的输出上、在将其加到子层输入并进行归一化之前应用丢弃 [27]。此外，我们还在编码器和解码器堆栈中对嵌入与位置编码之和和应用丢弃。对于基础模型，我们使用的比率为 $P_{drop} = 0.1$ 。

Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [15]	23.75			
Deep-Att + PosUnk [32]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [31]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [8]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [26]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [32]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [31]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [8]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (big)	28.4	41.0	$2.3 \cdot 10^{19}$	

Label Smoothing During training, we employed label smoothing of value $\epsilon_{ls} = 0.1$ [30]. This hurts perplexity, as the model learns to be more unsure, but improves accuracy and BLEU score.

6 Results

6.1 Machine Translation

On the WMT 2014 English-to-German translation task, the big transformer model (Transformer (big) in Table 2) outperforms the best previously reported models (including ensembles) by more than 2.0 BLEU, establishing a new state-of-the-art BLEU score of 28.4. The configuration of this model is listed in the bottom line of Table 3. Training took 3.5 days on 8 P100 GPUs. Even our base model surpasses all previously published models and ensembles, at a fraction of the training cost of any of the competitive models.

On the WMT 2014 English-to-French translation task, our big model achieves a BLEU score of 41.0, outperforming all of the previously published single models, at less than 1/4 the training cost of the previous state-of-the-art model. The Transformer (big) model trained for English-to-French used dropout rate $P_{drop} = 0.1$, instead of 0.3.

For the base models, we used a single model obtained by averaging the last 5 checkpoints, which were written at 10-minute intervals. For the big models, we averaged the last 20 checkpoints. We used beam search with a beam size of 4 and length penalty $\alpha = 0.6$ [31]. These hyperparameters were chosen after experimentation on the development set. We set the maximum output length during inference to input length + 50, but terminate early when possible [31].

Table 2 summarizes our results and compares our translation quality and training costs to other model architectures from the literature. We estimate the number of floating point operations used to train a model by multiplying the training time, the number of GPUs used, and an estimate of the sustained single-precision floating-point capacity of each GPU ⁵.

6.2 Model Variations

To evaluate the importance of different components of the Transformer, we varied our base model in different ways, measuring the change in performance on English-to-German translation on the development set, newstest2013. We used beam search as described in the previous section, but no checkpoint averaging. We present these results in Table 3.

In Table 3 rows (A), we vary the number of attention heads and the attention key and value dimensions, keeping the amount of computation constant, as described in Section 3.2.2. While single-head attention is 0.9 BLEU worse than the best setting, quality also drops off with too many heads.

⁵We used values of 2.8, 3.7, 6.0 and 9.5 TFLOPS for K80, K40, M40 and P100, respectively.

表2: Transformer在英语到德语和英语到法语newstest2014测试中, 以极低的训练成本超越了以往最先进模型的BLEU分数。

模型	BLEU		训练成本 (浮点运算次数)	
	英语-德语	英语-法语	英语-德语	英语-法语
ByteNet [15]	23.75			
Deep-Att + PosUnk [32]		39.2		$1.0 \cdot 10^{20}$
谷歌神经机器翻译 + 强化学习 [31]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [8]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [26]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk 集成 [32]		40.4		$8.0 \cdot 10^{20}$
谷歌神经机器翻译 + 强化学习 集成 [31]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S 集成 [8]	26.36	41.29	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (基础模型)	27.3	38.1	$3.3 \cdot 10^{18}$	
Transformer (大型)	28.4	41.0	$2.3 \cdot 10^{19}$	

标签平滑在训练期间, 我们采用了值为 $\epsilon_{ls} = 0.1$ [30]的标签平滑。这会损害困惑度, 因为模型学会变得更加不确定, 但能提高准确率和BLEU分数。

6 结果

6.1 机器翻译

在WMT 2014英语到德语翻译任务上, 大型Transformer模型(表2中的Transformer (big))比之前报告的最佳模型(包括集成)高出超过2.0 BLEU, 确立了28.4这一新的最先进BLEU分数。该模型的配置列于表3的最后一行。训练在8块P100 GPU上耗时3.5天。即便是我们的基础模型也超越了所有先前发表的模型和集成, 而训练成本仅为任何竞争模型的几分之一。

在WMT 2014英语到法语翻译任务上, 我们的大模型取得了41.0的BLEU分数, 超越了所有先前发表的单模型, 而训练成本不到先前最先进模型的 1/4。用于英语到法语的Transformer (big)模型训练使用了丢弃率 $P_{drop} = 0.1$, 而非0.3。

对于基础模型, 我们使用了通过对最后5个检查点进行平均得到的单一模型, 这些检查点每隔10分钟保存一次。对于大模型, 我们平均了最后20个检查点。我们使用束大小为4的束搜索和长度惩罚 $\alpha = 0.6$ [31]。这些超参数是在开发集上进行实验后选定的。我们在推理时将最大输出长度设置为输入长度 + 50, 但在可能的情况下提前终止 [31]。表2总结了我们的结果, 并将我们的翻译质量和训练成本与文献中的其他模型架构进行了比较。我们通过将训练时间、使用的GPU数量以及每块GPU的持续单精度浮点能力估计值 ⁵相乘, 来估算训练一个模型所用的浮点运算次数。

6.2 模型变体

为了评估Transformer不同组件的重要性, 我们以不同方式修改了基础模型, 测量在开发集newstest2013上的英德翻译性能变化。我们使用了前一节所述的束搜索, 但没有使用检查点平均。我们在表3中呈现了这些结果。

在表3的(A)行中, 我们按照第3.2.2节的描述, 在保持计算量不变的情况下, 改变注意力头的数量以及注意力键和值维度。虽然单头注意力的BLEU分数比最佳设置低0.9, 但头数过多时质量也会下降。

⁵我们使用了 2.8、3.7、6.0 和 9.5 TFLOPS 分别对应 K80、K40、M40 和 P100。

Table 3: Variations on the Transformer architecture. Unlisted values are identical to those of the base model. All metrics are on the English-to-German translation development set, newstest2013. Listed perplexities are per-wordpiece, according to our byte-pair encoding, and should not be compared to per-word perplexities.

	N	d_{model}	d_{ff}	h	d_k	d_v	P_{drop}	ϵ_{ts}	train steps	PPL (dev)	BLEU (dev)	params $\times 10^6$	
base	6	512	2048	8	64	64	0.1	0.1	100K	4.92	25.8	65	
(A)					1	512	512				5.29	24.9	
					4	128	128				5.00	25.5	
					16	32	32				4.91	25.8	
					32	16	16				5.01	25.4	
(B)					16					5.16	25.1	58	
					32					5.01	25.4	60	
(C)	2									6.11	23.7	36	
	4									5.19	25.3	50	
	8									4.88	25.5	80	
		256				32	32				5.75	24.5	28
		1024				128	128				4.66	26.0	168
			1024								5.12	25.4	53
			4096								4.75	26.2	90
(D)							0.0				5.77	24.6	
							0.2				4.95	25.5	
								0.0		4.67	25.3		
								0.2		5.47	25.7		
(E)	positional embedding instead of sinusoids									4.92	25.7		
big	6	1024	4096	16				0.3	300K	4.33	26.4	213	

In Table 3 rows (B), we observe that reducing the attention key size d_k hurts model quality. This suggests that determining compatibility is not easy and that a more sophisticated compatibility function than dot product may be beneficial. We further observe in rows (C) and (D) that, as expected, bigger models are better, and dropout is very helpful in avoiding over-fitting. In row (E) we replace our sinusoidal positional encoding with learned positional embeddings [8], and observe nearly identical results to the base model.

7 Conclusion

In this work, we presented the Transformer, the first sequence transduction model based entirely on attention, replacing the recurrent layers most commonly used in encoder-decoder architectures with multi-headed self-attention.

For translation tasks, the Transformer can be trained significantly faster than architectures based on recurrent or convolutional layers. On both WMT 2014 English-to-German and WMT 2014 English-to-French translation tasks, we achieve a new state of the art. In the former task our best model outperforms even all previously reported ensembles.

We are excited about the future of attention-based models and plan to apply them to other tasks. We plan to extend the Transformer to problems involving input and output modalities other than text and to investigate local, restricted attention mechanisms to efficiently handle large inputs and outputs such as images, audio and video. Making generation less sequential is another research goals of ours.

The code we used to train and evaluate our models is available at <https://github.com/tensorflow/tensor2tensor>.

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表3: Transformer架构的变体。未列出的值与基础模型相同。所有指标均基于英语到德语翻译开发集newstest2013。所列困惑度为每词片困惑度，依据我们的字节对编码计算，不应与每词困惑度进行比较。

	N	$d_{\text{模型}}$	d_{ff}	h	d_k	d_v	P_{drop}	ϵ_{ls}	训练 训练步数	PPL (开发集)	BLEU (开发)	参数量 $\times 10^6$	
base	6	512	2048	8	64	64	0.1	0.1	100K	4.92	25.8	65	
(A)					1	512	512				5.29	24.9	
					4	128	128				5.00	25.5	
					16	32	32				4.91	25.8	
					32	16	16				5.01	25.4	
(B)					16					5.16	25.1	58	
					32					5.01	25.4	60	
(C)	2									6.11	23.7	36	
	4									5.19	25.3	50	
	8									4.88	25.5	80	
		256			32	32				5.75	24.5	28	
		1024			128	128				4.66	26.0	168	
			1024								5.12	25.4	53
			4096								4.75	26.2	90
(D)									0.0	5.77	24.6		
									0.2	4.95	25.5		
									0.0	4.67	25.3		
									0.2	5.47	25.7		
(E)	位置嵌入而非正弦曲线									4.92	25.7		
big	6	1024	4096	16				0.3	300K	4.33	26.4	213	

在表3的（B）行中，我们观察到减小注意力键大小 d_k 会损害模型质量。这表明确定兼容性并不容易，而且比点积更复杂的兼容性函数可能更有益。我们在（C）行和（D）行中进一步观察到，正如预期的那样，更大的模型更好，并且Dropout在避免过拟合方面非常有帮助。在（E）行中，我们用学习式位置嵌入 [8],替换正弦位置编码，并观察到与基础模型几乎相同的结果。

7 结论

在这项工作中，我们提出了 Transformer，这是第一个完全基于注意力的序列转导模型，用多头自注意力取代了编码器-解码器架构中最常用的循环层。

对于翻译任务，Transformer 的训练速度可以显著快于基于循环层或卷积层的架构。在 WMT 2014 英语到德语和 WMT 2014 英语到法语翻译任务上，我们都达到了新的最优水平。在前一项任务中，我们的最佳模型甚至超越了所有此前报告的集成模型。

我们对基于注意力的模型的将来感到兴奋，并计划将其应用于其他任务。我们计划将 Transformer 扩展到涉及文本以外的输入输出模态的问题，并研究局部、受限的注意力机制，以高效处理图像、音频和视频等大型输入和输出。让生成过程减少顺序性，也是我们的另一个研究目标。

我们用于训练和评估模型的代码可在 <https://github.com/tensorflow/tensor2tensor> 获取。

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